

XIAMEN UNIVERSITY

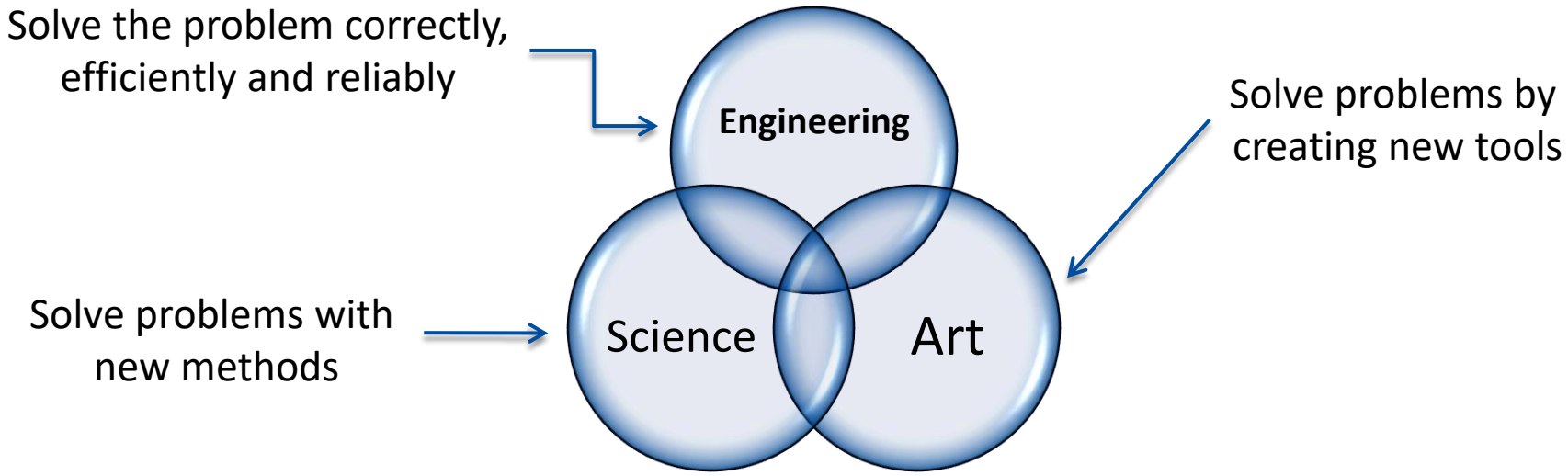
202109 COM02 ROGRAMMING IN LANGUAGE C

Course Information

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What is this class about?

- This course is an introduction to programming.
- What is programming?
  - solve real-world problems with a computer



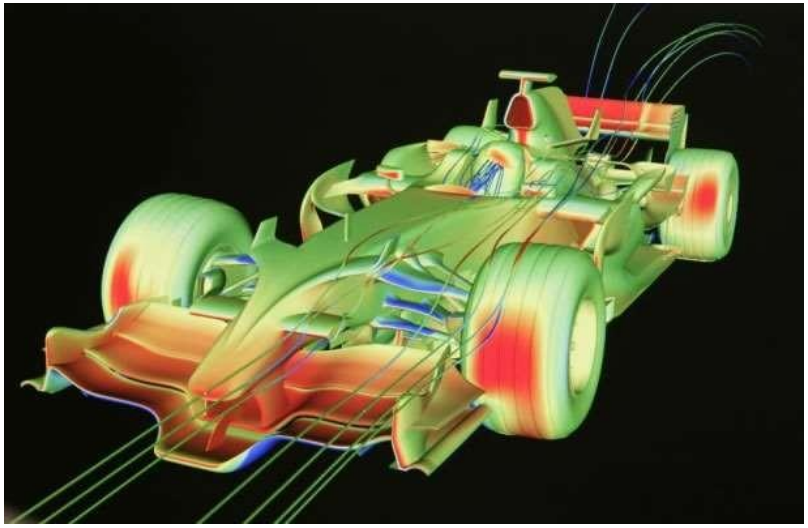
Application examples

- Robotics



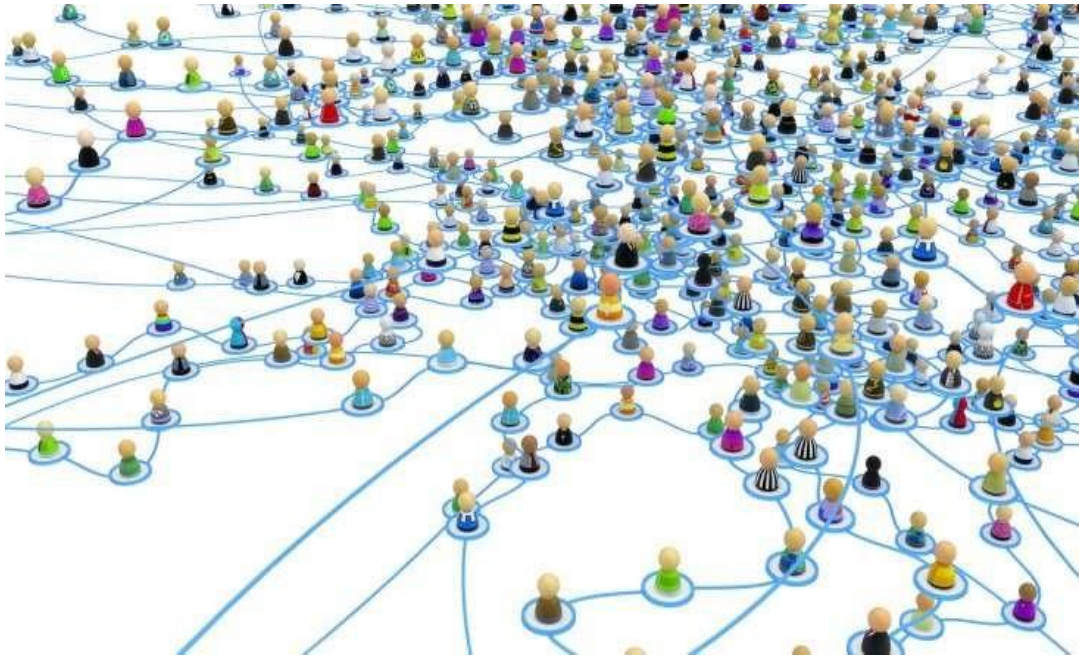
Application examples

- Scientific computing & simulations
  - Aerodynamics



Application examples

- Social network, Big data, Data Mining, etc.



Semester and Year Offered

- 04 semester or 09 semester
- Y1

Credit Value

- 3

Programme Educational Objectives (PEOs)

PEO1	Professionalism-Graduates will establish themselves as practicing professionals in the field of Electrical and Electronics Engineering or related fields.
PEO2	Problem analysis and Development of Solutions-Identify, formulate, conduct research literature and analyse complex engineering problems and design solutions that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PEO3	Continuous Personal Development-Graduates will engage in lifelong pursuit of knowledge and interdisciplinary learning appropriate for industrial and academic careers.

Programme Learning Outcomes (PLOs)

PLO1	Engineering Knowledge - Apply knowledge of mathematics, science, electrical & electronic engineering fundamentals and engineering specialization principles to define and utilize engineering procedures, processes, systems and methodologies.
PLO2	Problem Analysis - Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PLO3	Design/Development of Solutions - Design solutions for complex engineering problems and systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

Course Objectives

- To provide foundation of the computer programming in using conditional and iterative structures, functional decomposition, and basic parallelization techniques.
- To provide engineering students with basic knowledge of programming in C.
- To help the students to learn to write some commonly used algorithms and subroutines, such as find the maximum from an array, bubble sorting, string manipulating, pointer operation, etc.
- To prepare students for their future work in the development of large scale software systems.

Learning Outcomes

- Understand and find the output of simple C programs that incorporate different types of variables, expressions (arithmetic and logical), selection, and iteration;
- Design and implement simple programs using basic syntax of C language such as assignment, expressions, selection, and iterations.
- Utilize the modular features of the language.
- Develop a computer algorithm for the engineering problem.

Course Outline

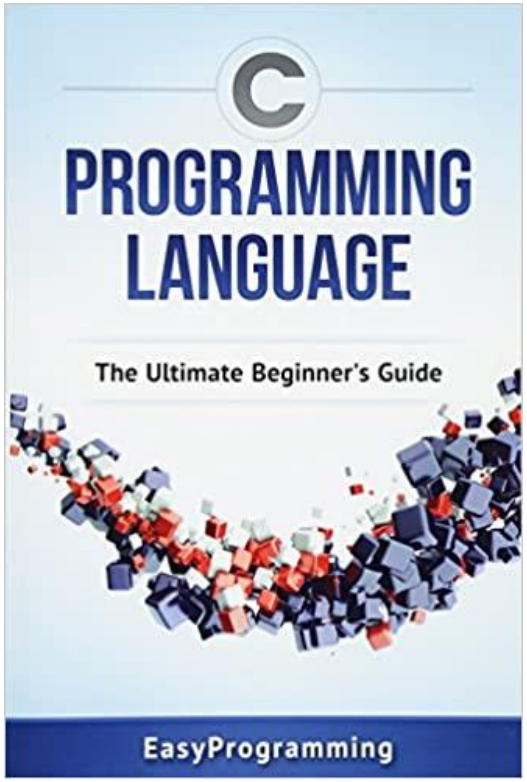
- 1. Introduction
- 2. Types, Operators and Expressions
- 3. Control Flow
- 4. Functions and Program Structure
- 5. Pointers and Array
- 6. Structures and Strings
- 7. Input and Output

Course Information

- Moodle System
  - http://l.xmu.edu.my/
  - Course Short Name: 2029/04-COM102 Programming Language C
  - Enrolment key: COM102USEFUL

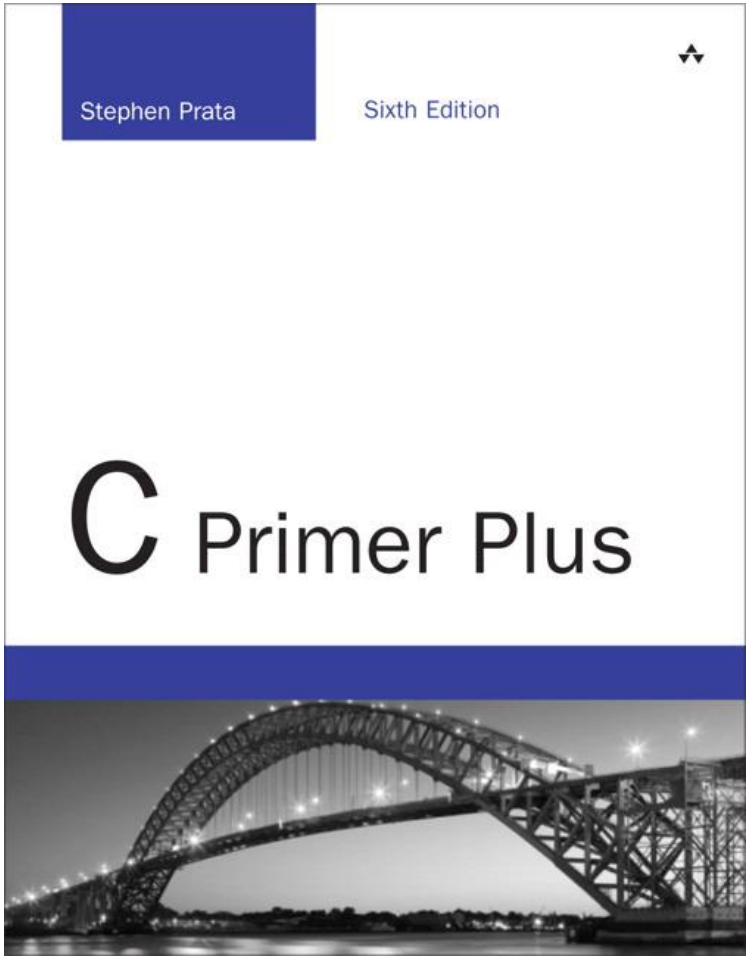
Required References

- Easy Programming, "C Programming Language: The Ultimate Beginner's Guide," CreateSpace Independent Publishing Platform (November 12, 2016).



Required References

- Stephan Prata, C primer plus 6th edition. Pearson, USA. (2014)



Further Readings

- Herbert Schildt, C The Complete Reference, Fourth Edition, ISBN-13: 978-0072121247, April 26th 2000, The McGraw-Hill Companies.

Course Information

- Assessment Methods and Types:

Coursework	**50%**
• **Quizzes**	30%
• **Assignment**	20%
Final Examination	50%
Total	100%

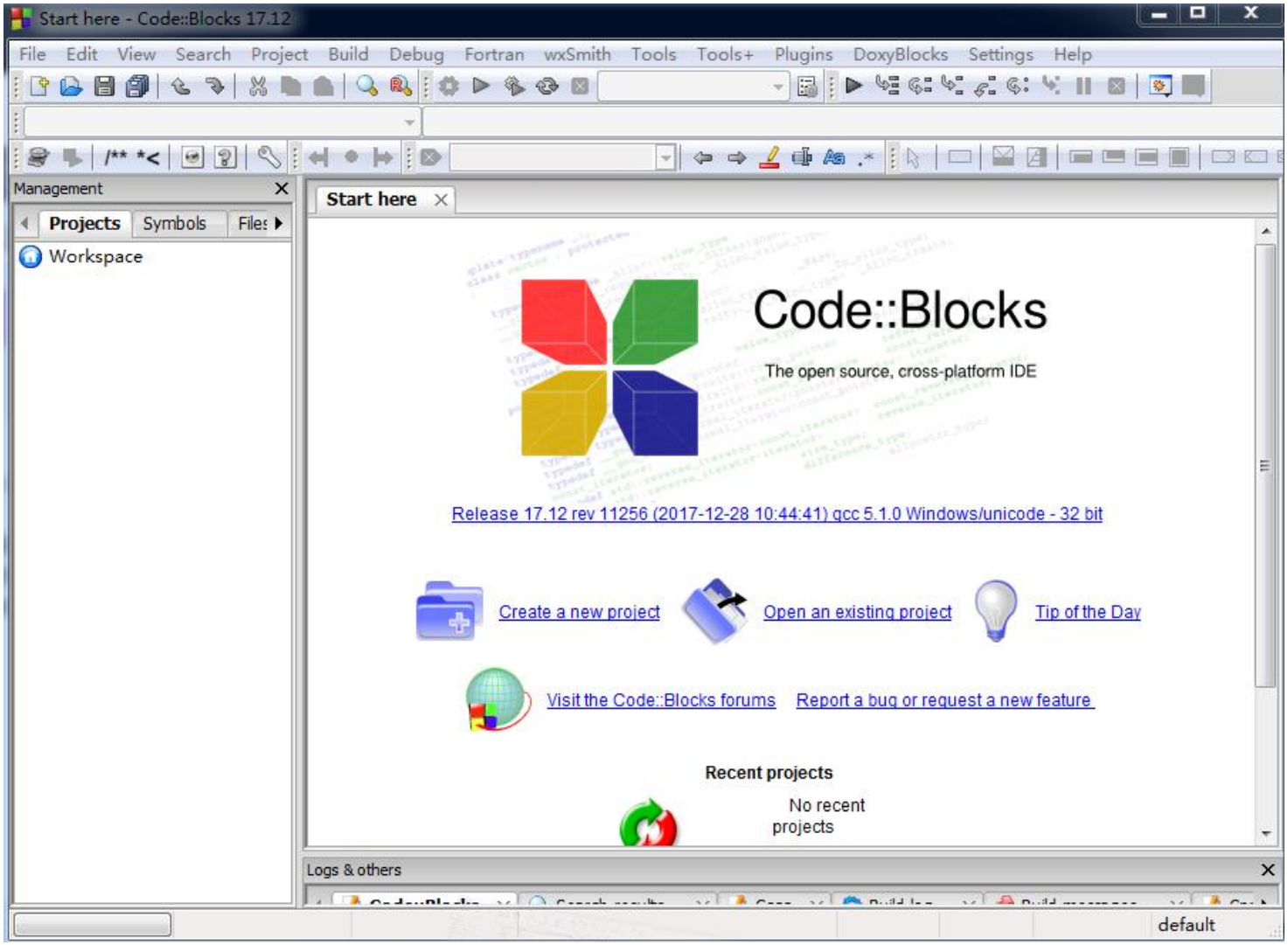
Grading System

Grading Scales and grades applicable to ALL intakes effective from Academic Session 2017/09:

Grading System for Academic Session 2017/09 onwards			
Marks	**Grade**	**Grade Points**	**Description**
85 – 100	A	4.00	Distinction
80 – 84	A-	3.70	Distinction
75 – 79	B+	3.30	Good
70 – 74	B	3.00	Good
65 – 69	B-	2.70	Good
60 – 64	C+	2.30	Pass
55 – 59	C	2.00	Pass
50 - 54	C-	1.70	Conditional Pass
45 - 49	D+	1.30	Fail
40 - 44	D	1.00	Fail
≤ 39	F	0.00	Fail

Table 1: Grading System Effective from Academic Session 201709

Codeblocks IDE



Codeblocks Download Link:

- http://www.codeblocks.org/downloads/26

Assignment

- Late policy:
  - Before deadline: 100%
  - 24hour Late: 75%
  - 24hour - 48hour Late: 50%
  - > 48hour: 0%

My Expectations - Attendance

- Attendance is required by Ministry of Higher Education Malaysia
- Attendance sheet should be signed on each class, including lab sessions
- Sick leave and other types leaves should go through standard procedures
- Students will be barred from sitting final exam if their attendance rate is <80% of each course).

Consultation Time

- 4-hour per week
  - Tuesday, 12pm-14pm
  - Friday, 12pm-14pm

Contact me

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